

Bike Sales Profitability Analysis

Excel data-quality audit, analysis layer & interactive dashboard — project summary

EXECUTIVE SUMMARY (STAKEHOLDER VERSION)

We analyzed ~113,000 sales transactions (2011–2016) to identify where the business makes and loses money. A data-quality audit revealed that roughly half the dataset (years 2012, 2015, 2016) was fabricated — duplicated from earlier years with altered quantities — along with 1,000 duplicate records. All findings below are based exclusively on the verified 56,000 genuine transactions.

Key findings:

- The business generated \$38.1M revenue at a 38.3% profit margin on verified data.
- Discounting is the biggest profit leak. 95% of transactions are discounted (avg 10.4%), and margin falls steadily from 51% on full-price sales to 31% in the deepest discount band. The >15% discount band alone covers \$11M+ of revenue at the weakest margins — tightening discount policy is the clearest opportunity.
- Helmets are the most profitable products in the portfolio (~59% margin, outearning the bikes themselves) — a scale-up candidate. Classic jerseys sit at 10–13% margin with above-average discounts — reprice, renegotiate cost, or cut.
- Canada is the margin benchmark: 47.4% margin with almost no discounting (1%), proving the product sells without heavy promotion.

Deliverables: a cleaned, audit-ready dataset; a formula-driven analysis workbook (category, country, product, discount-band, and trend views); and an interactive dashboard with country-level filtering.

PROJECT SUMMARY (RECRUITER VERSION)

Built an end-to-end Excel profitability analysis of 113K sales transactions, working with an AI assistant for data profiling and build-out while directing the approach and validating results. Key contribution: caught that ~50% of the dataset was fabricated — duplicate years disguised with jittered quantities — and proved it with a leap-day test (the 2012/2016 "leap years" contained zero Feb-29 sales) plus structural row-matching (99%+ pairing). Flagged rather than deleted the fake data for auditability, then delivered a cleaned table, a fully formula-driven analysis layer (margins by category/country/product, discount-erosion banding, like-for-like trends), and an interactive dashboard with a country filter. Headline insight: discounting erodes margin monotonically from 51% to 31%, while the highest-margin products (helmets, ~59%) outearn the flagship bikes.

WORKBOOK STRUCTURE

- Sales — raw data, untouched (113,037 rows)
- Sales_clean — deduplicated table with Order_ID key, derived discount/margin columns, and Is_Original authenticity flag (112,036 rows)
- Analysis — formula-driven: KPIs, category/country/sub-category profitability, real-year trends, discount-erosion bands, top & bottom 10 products
- Products — per-product profitability detail for all 130 products with profit and margin rankings
- Dashboard — KPI cards, four charts, interactive country filter

Bike Sales Profitability Analysis

113K transactions · data forensics · P&L analysis · interactive Excel dashboard — built with AI assistance, directed & validated by the analyst

REVENUE	PROFIT	MARGIN	AVG DISCOUNT	UNITS SOLD	VERIFIED SALES
\$38M	\$14.6M	38.3%	10.4%	673,629	56,017

THE CATCH: HALF THE DATASET WAS FAKE

A data-quality audit revealed that years 2012, 2015 and 2016 — 50% of all rows — were fabricated: copied from 2011, 2013 and 2014 with shifted dates and altered quantities. Proven two ways: (1) the leap-day test — 2012 and 2016 are leap years averaging 140+ sales per day, yet contain zero Feb-29 transactions — impossible for real data, inevitable for copied dates; (2) structural row-matching — 99%+ of fabricated rows pair one-to-one with a source-year row. Fake data was flagged, not deleted, keeping the workbook auditable. Every number in this report uses verified data only.

WHAT THE VERIFIED DATA SAYS

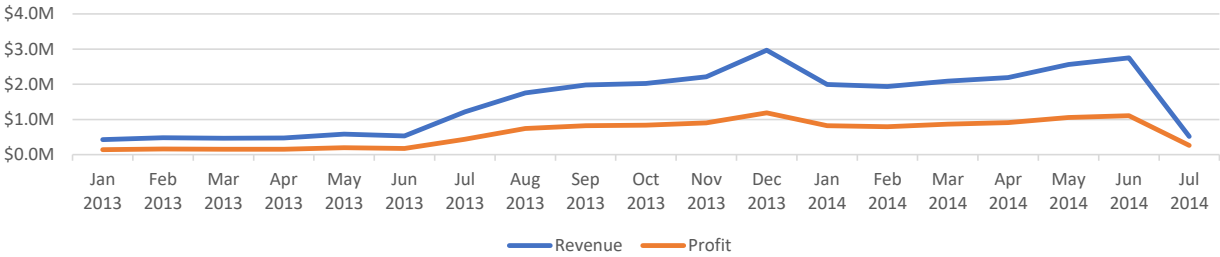
- Discounting is the #1 profit leak: 95% of sales are discounted; margin erodes step-by-step from 51% (full price) to 31% (>15% off). Tightening discount policy is the clearest win.
- Helmets out-earn the bikes: Sport-100 helmets hold 3 of the top 4 profit spots at ~59% margin — the scale-up candidate hiding in the accessories aisle.
- Classic jerseys are the fix-or-cut zone: 10–13% margins with above-average discounts.
- Canada proves the ceiling: 47.4% margin with just 1% average discount — the product sells without heavy promotion.

HOW IT WAS BUILT (HUMAN + AI)

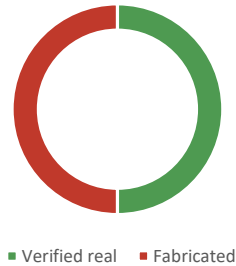
The analyst directed an AI assistant through full-dataset profiling (all 113K rows scanned, not sampled), removal of 1,000 duplicates, authenticity flagging, and a fully formula-driven analysis layer — every figure traceable to a live Excel formula — plus an interactive dashboard with country filtering. The analyst set the approach, challenged intermediate results and validated every output.

Workbook: Sales (raw) → Sales_clean → Analysis → Products → Dashboard → README.

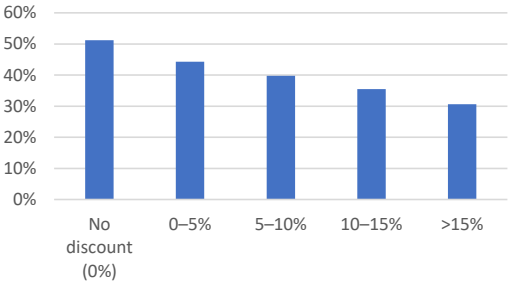
Monthly Revenue & Profit (verified, Jan 2013 – Jul 2014)



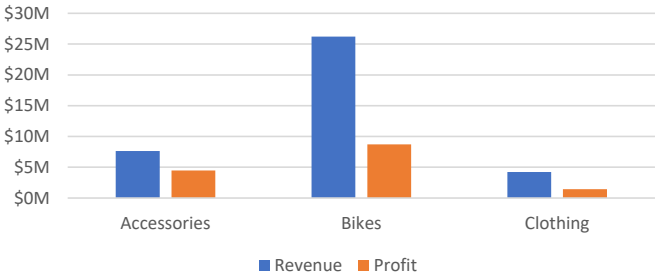
113,036 rows: real vs fabricated



Margin % by Discount Band



Category: Revenue vs Profit



Top 10 Products by Profit

